

Sebastian Jitaru

Github Profile | sjitaru29@gmail.com

EDUCATION

MSc, Artificial Intelligence

Sept 2024 – Aug. 2026

University of Groningen, Netherlands

- **Machine Learning/ Robotics Track** Coursework included Deep Learning, Pattern Recognition, Natural Language Processing with Deep Learning, Deep Reinforcement Learning, Multi-Agent Reinforcement Learning, Trustworthy & Explainable AI, Computer Vision, Cognitive Robotics, Advanced Machine Learning, Machine Learning Operations (MLOPs)

BSc, Computer Engineering

September 2020 – July 2024

University of Lleida, Spain

- **Cybersecurity/ Software development Track** Coursework included Algorithms & Data-Structures, Object Oriented Programming, Calculus, Statistics, Computer Architecture, Computational Logic, Physics, Systems Administration, Communication Networks, Software Architectures, Mobile Applications, Advanced Networks and Communications, Distributed Computing, Application Security, Cryptography.

EXPERIENCE

AI Engineer, Prioticket

January 2025 – Present

Amsterdam, Netherlands

- Designed a dynamic pricing module using an ensemble of gradient boosting and regularized regression models to predict optimal price per product per client. Built the MLOps pipeline with monitoring of feature and prediction distributions to detect data drift and distribution shifts, automated model evaluation against baseline metrics, and continuous retraining and deployment triggered upon performance degradation.
- Built a RAG pipeline with hybrid sparse-dense retrieval, reciprocal rank fusion, and cross-encoder re-ranking over a centralized knowledge base, reducing manual completion time by 60%.
- Designed an automated data aggregation pipeline for client onboarding using Named Entity Recognition, dense embedding similarity for fuzzy deduplication, and rule-based classification to produce clean, structured client profiles.

Software Engineer, Prioticket

July 2024 – December 2024

Amsterdam, Netherlands

- Engineered and maintained full-stack features across multiple codebases using TypeScript, Angular, NestJS and Express.js with async/event-driven patterns, integrated with BigQuery and CloudSQL. Designed idempotent REST APIs with proper error propagation and request validation, optimizing response times monitored using tools like NewRelic or Greylog.
- Managed CI/CD pipelines using GitLab for various services (e.g., Reporting Frontend and Backend), deploying to Google Cloud Kubernetes clusters with automated scaling and monitoring, achieving a 99.9% annual uptime SLA.
- Implemented multi-tenant reporting features with role-based access control and tenant-isolated query scoping, export functionalities for enterprise clients, and financial system integrations with external APIs (Twinfield, Oracle Finance), following Agile SCRUM practices.

Cybersecurity Analyst, Semic

June 2023 – June 2024

Lleida, Spain

- Deployed and maintained endpoint protection and EDR solutions (CrowdStrike, Kaspersky, Bitdefender) for public and private sector clients, such as the Hospital of Barcelona-Granollers or the Government of Andorra.
- Investigated security incidents such as ransomware intrusions targeting SMEs in Lleida Catalonia, performing log and network traffic analysis with CrowdStrike and Kaspersky, and led internal security audits and vulnerability assessments using OWASP methodologies and tools like Burp Suite.
- Administered Fortinet firewalls (configuration, monitoring, and policy enforcement), managed phishing simulation campaigns via cloned web assets, deployed honeypots for threat intelligence, and managed server environments (Windows/Linux), including troubleshooting connectivity and system-level issues.

Machine Learning Engineer - Ruxailab

Feb 2024 – July 2024

Internship, Brazil

- Designed and trained a custom CNN on the FER dataset for real-time video emotion analysis achieving 72% per-frame accuracy. Built the serving infrastructure with FastAPI (async endpoints, batch inference) and Vue.js, containerized with Docker, and deployed on GCP with Firebase for auth and persistence.

Research projects

- **Do LLMs Think Too Much? Adaptive Chain-of-Thought Length Reduction via Latent State Prediction.**
Ongoing research project at University of Groningen - Jeroen Niemendal, Viktor Vesely, Sebastian Jitaru
 - Proposed a novel CoT Length Reduction method via classification of the LLM internal hidden state, where the model dynamically predicts the minimum amount of tokens required to generate a trajectory with an accurate solution.
- **Liquid Ensembles: Differentiable Liquid Democracy for Decentralized Model Specialization.**
Ongoing research project at University of Groningen - Viktor Vesely, Sebastian Jitaru.
 - Proposed a deep ensemble method where models dynamically delegate decisions to other models via fully differentiable mechanisms, enabling emergent task specialization.
 - Planning to publish at Transactions on Machine Learning Research.
- **Guided molecule selection using Chemical Foundational Models for Peptide Design.**
Ongoing Master Thesis at University of Groningen & Traffikgene company - Sebastian Jitaru
 - Built an end-to-end ML system from prototype to deployment: data preprocessing and validation pipeline, model training, and large-scale batch inference over a candidate pool of millions, in Python (PyTorch, HuggingFace Transformers, scikit-learn) on HPC (SLURM). Fine-tuned a transformer foundation model with LoRA for parameter-efficient adaptation, and ran domain-adaptive continued pretraining of the encoder backbone. Built a heterogeneous ensemble for prediction with out-of-distribution guardrails to flag low-confidence outputs, reducing wasted downstream cost by prioritizing only reliable, high-value candidates.
 - Interpreted the CLM backbone by applying Sparse Autoencoders and Max Affine Spline Operators theory to Chemistry foundational models to analyse their internal representation and automatically classify the latents as human interpretable features.
- **Production ML System for Issue Classification.**
GitHub - University of Groningen
 - Built an end-to-end ML system for multi-label issue classification using a fine-tuned BERT model. Implemented the full MLOps lifecycle: DVC for data versioning and reproducible pipelines, MLflow with MinIO for experiment tracking and artifact storage, Pandera for schema validation, and FastAPI with async endpoints and batch inference for model serving.
 - Deployed a complete observability stack (Prometheus, Grafana, Loki, Promtail) with hybrid push/pull monitoring, stored predictions in MongoDB, containerized all services with Docker Compose, and automated the build-test-train-deploy pipeline via GitLab CI/CD with remote VM deployment.

CERTIFICATES

- Cambridge Proficiency Exam (CAE) C2
- Crowdstrike
- Kaspersky
- Btodefender
- Google Cloud

SKILLS

Security: CrowdStrike, Kaspersky, Bitdefender, Fortinet, Burp Suite, Nmap, SIEM (Splunk, QRadar), MITRE ATT&CK, YARA, Sigma, OWASP, EDR, threat intelligence, incident response, vulnerability assessment, network administration (VLANs, routing, switching)

ML Frameworks: PyTorch, JAX, vLLM, Scikit-learn, LangChain

MLOps & Infra: MLflow, DVC, Pandera, Docker, Kubernetes, Grafana, Prometheus, Loki, GitLab CI/CD, Vertex AI, GCP, AWS, NewRelic, Graylog, Cloudflare

HPC & Compute: SLURM, CUDA, multi-node GPU cluster training

Data & Storage: SQL, MongoDB, BigQuery, Elasticsearch, Redis, MinIO

Backend: Python, TypeScript, JavaScript, NestJS, FastAPI, Express.js, Java

Frontend: Angular.js, Next.js, React, CSS, HTML

Architecture & Practices: event-driven & async architectures, stream processing, REST API design, replication & partitioning strategies, OOP, SOLID, design patterns, Git, Linux